

Bachelor of Science in Civil Engineering

Year of study: 2nd Year; 2nd Semester

Course Title: CIVIL ENGINEERING DRAWING AND
DESIGN

Course Code: FCC3242

TUTORIAL TWO

SECTIONAL VIEWS

(a) Truncated cylinder

A solid cylinder whose base is 60mm in diameter and has a height of 80mm is initially positioned in such way that its axis is vertical. A quadrant cutting plane, which is perpendicular to the front picture plane, cuts the axis of the cylinder at a point 50mm above the base as shown in Fig.1. The remaining portion below the cutting plane is revolved in the anti-clockwise direction until the axis makes an angle of 45° to the horizontal.

Using a suitable scale, draw:

- i. Front and top views of the truncated cylinder in the vertical position.
- ii. Front, left side and the top views of the truncated cylinder in the revolved position.

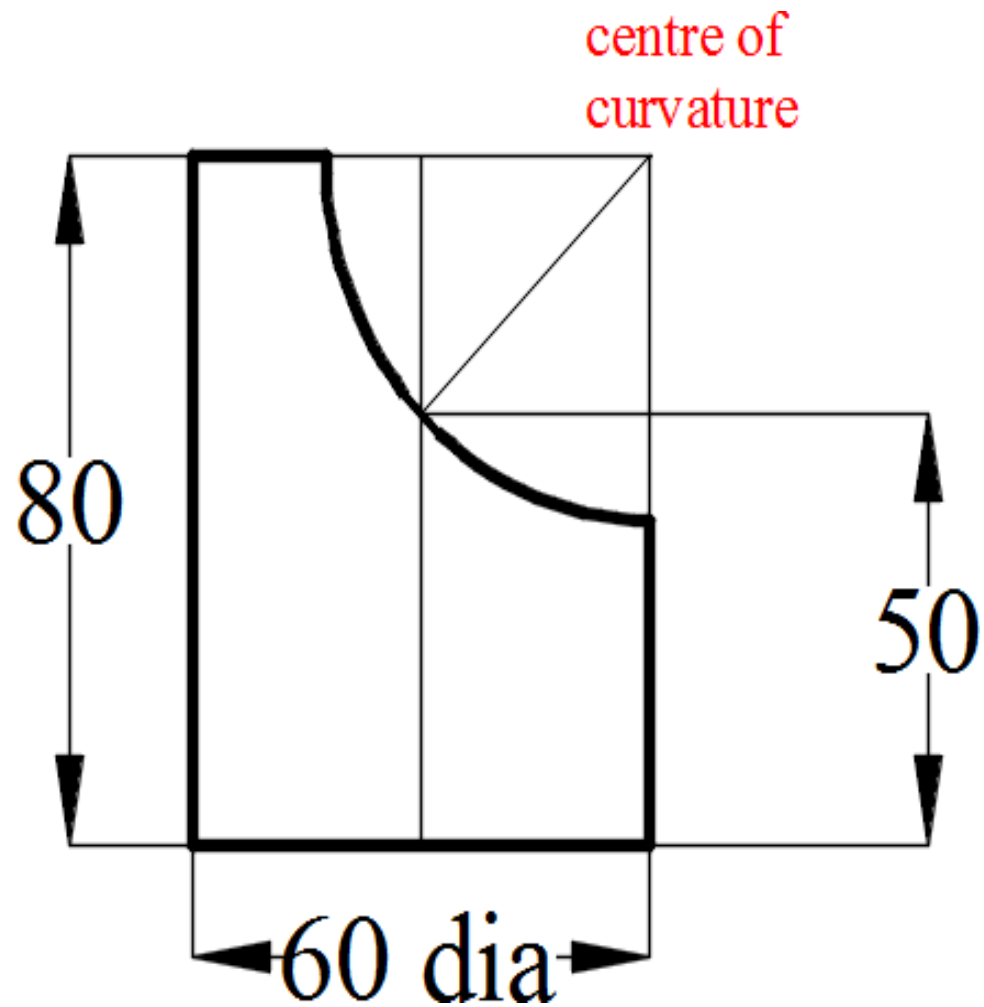


Fig. 1

SUGGESTED PROCEDURE FOR DRAWING (a)

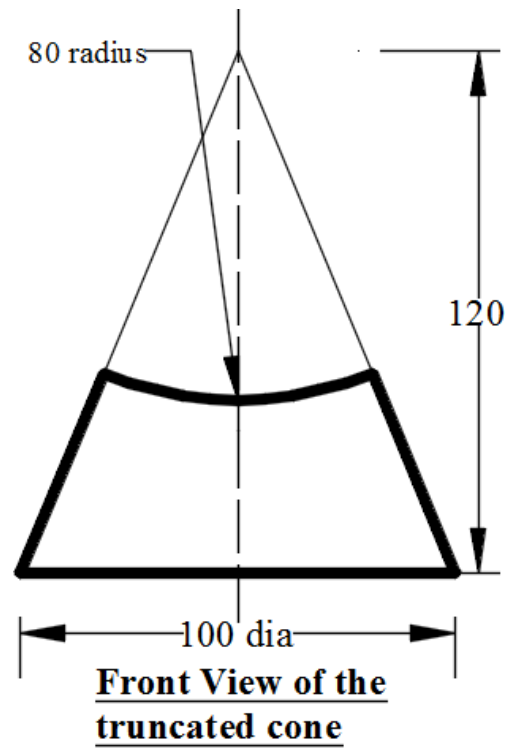
- Start by drawing the top view of a cylinder – which is just a circle
- Subdivide the circumference into 12 parts and identify them as 1 – 12 “corners” as in topic 1
- Project from these 12 points vertically upwards to find a horizontal line drawn 40mm above the circle
- Draw the front view of the cylinder which is just a rectangle; extend the projector line from its base to its top line. Identify the resultant corners
- Now it is time for drawing the curved cutting plane – noting carefully its centre of curvature; (visualize it a “glass piece of a cylinder” cutting the vertical cylinder in a horizontal direction from your eyes into the paper)
- All this time you are using construction lines – thin, continuous, sharp, visible)
- Identify the points where the vertical projectors cut the curved cutting plane, including the one at the top edge of the cylinder in question; you can use letters a, b, c, for this purpose
- Identify these points on the top view
- Thicken the visible edges on the two views
- You are now ready to manipulate the front view (complete with its corners) to any required orientation
- Recall the importance of using the top view in the initial position, to draw subsequent top views as was done in topic one

(b) Truncated cone

The figure below is the front view of a solid cone that has been truncated by a curved plane that is perpendicular to the front Picture Plane. The base of the cone is 100mm in diameter and its height is 120mm. The radius of the curved plane is 80mm.

Using first angle orthographic projection and choosing a scale of 1:1,

- (a) Draw the Front View, Left Side View and the Top View of the object. Identify clearly the cut edges.
- (b) The front view of the truncated cone is then tilted in the clockwise direction until the axis of the cone makes an angle of 60° to the horizontal. Draw the front, left side and the top views of this tilted position.

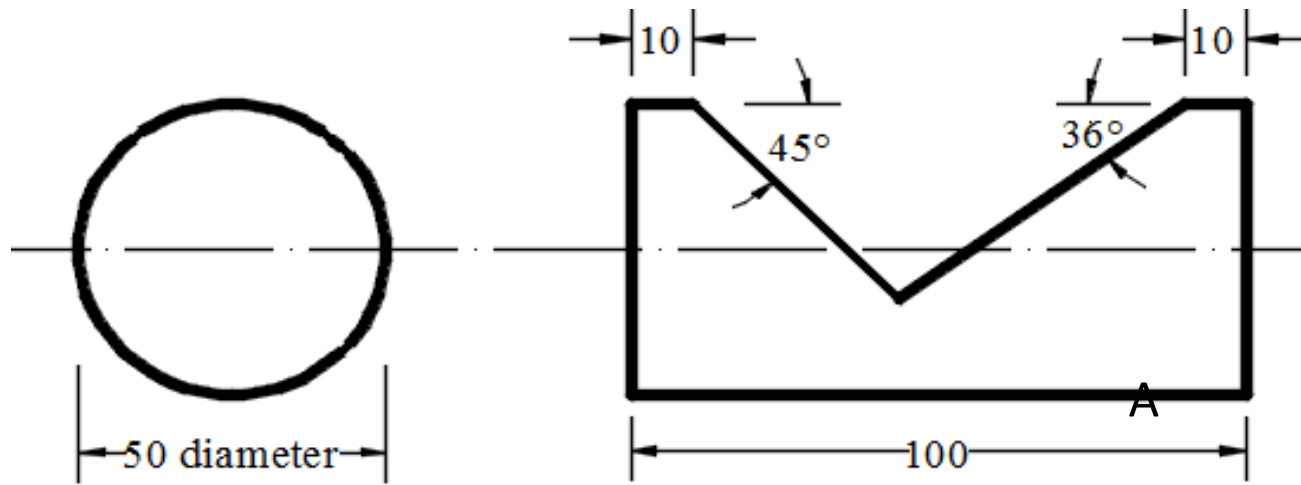


(c) Horizontal cylinder

The object shown in Figure Q1 is what remained of a solid horizontal cylinder that has been cut by two planes inclined as indicated in the Front View. The two cutting planes are perpendicular to the front picture plane. The length of the cylinder is 100mm while its cross-section has a diameter of 50mm.

Using first angle orthographic projection and drawing to a scale of 1:1,

- (a) Draw the Front View and the Top View of the object. Identify clearly the cut edges
- (b) The front is then tilted in the clockwise direction, about the position marked A, until the axis of the cylinder is vertical. Draw the Front view and the Right Side view.



RIGHT SIDE VIEW

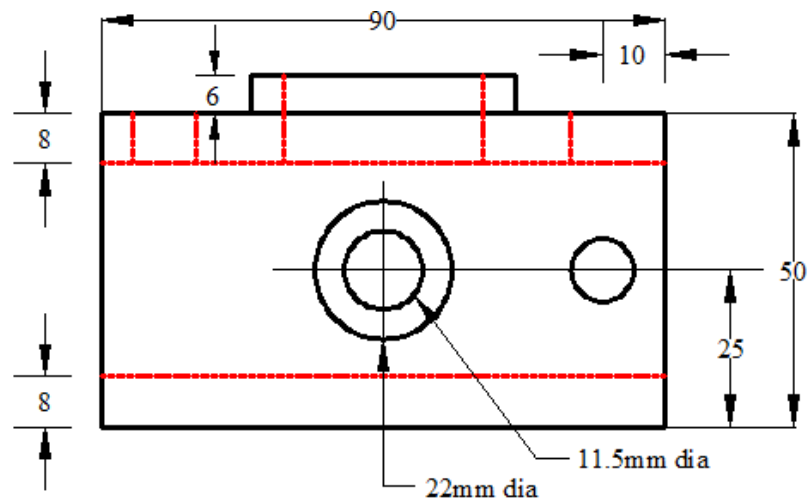
FRONT VIEW

(d) Metal sleeve

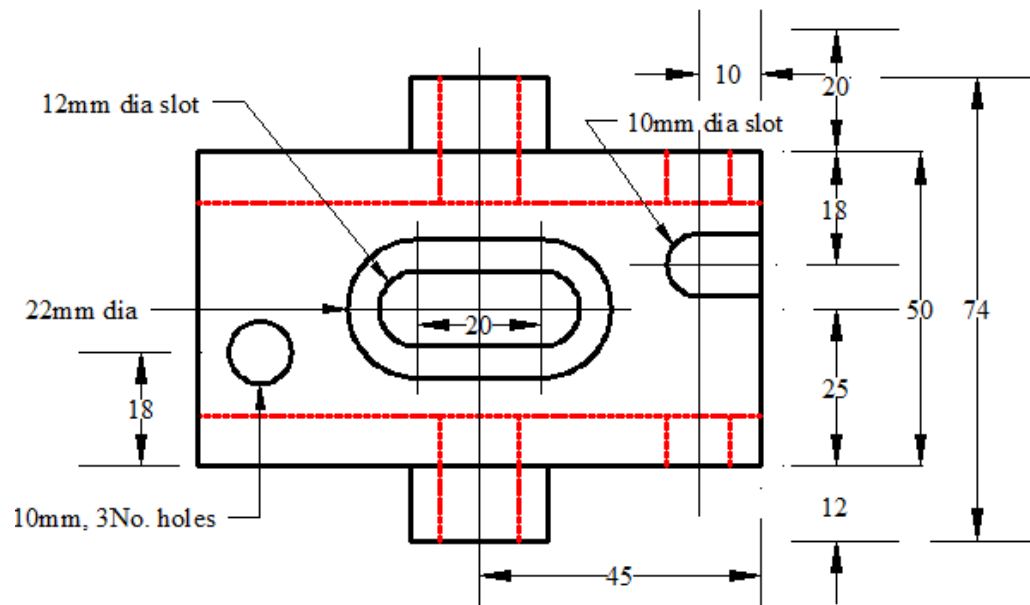
Below is the Front and Top Views of a metal sleeve. Using a scale of 1:1,

(a) Draw the Right Side View

(b) Choose and draw suitable sections to improve the clarity of the drawing



FRONT VIEW



TOP VIEW

Note that the red lines denote hidden details, to be drawn in the usual style: thick dashes

**TO BE SUBMITTED ON
MONDAY 17TH MARCH
2024 AT 2.00PM**